THE SCHEME THEORETIC IMAGE OF A PARTIAL SECTION IS AN ISOMORPHISM OVER THE DOMAIN OF THE SECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof uses the interaction of two same-chapter statements (commutation with opens and closedness of sections) to conclude an isomorphism; every step is written out.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Permanence of quasi-compactness for the section s. Lemma lemma-quasi-compact-permanence, used at proof lines 3393-3394. Makes s quasi-compact so that the scheme theoretic image behaves well on opens.
- The scheme theoretic image commutes with restriction to opens for quasi-compact morphisms. Lemma lemma-quasi-compact-scheme-theoretic-image, used at proof lines 3395-3397. Identifies Y' intersect f^{-1}(V) with the image of the restricted section.
- A section of a separated morphism is a closed immersion. Lemma lemma-section-immersion, used at proof lines 3398-3402. Turns the section of f^{-1}(V) -> V into a closed immersion.
- The scheme theoretic image of a closed immersion is the closed subspace itself. used at proof lines 3402-3404. Concludes that the image map over V is an isomorphism.

Source reference. Original lemma 03KS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4688–4688. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 082Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10269–10269. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4685–4685. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a separated morphism of algebraic spaces over S. Let $V \subset Y$ be an open subspace such that $V \to Y$ is quasi-compact. Let $s : V \to X$ be a morphism such that $f \circ s = id_V$. Let Y' be the scheme theoretic image of s. Then $Y' \to Y$ is an isomorphism over V.*

Proof. By Lemma 03KS the morphism $s : V \to X$ is quasi-compact. Hence the construction of the scheme theoretic image Y' of s commutes with restriction to opens by Lemma 082Z. In particular, we see that $Y' \cap f^{-1}(V)$ is the scheme theoretic image of a section of the separated morphism $f^{-1}(V) \to V$. Since a section of a separated morphism is a closed immersion (Lemma 03KP), we conclude that $Y' \cap f^{-1}(V) \to V$ is an isomorphism as desired. □